Research questions from the current analysis

Multimodal drivers + TerraMind embeddings for daily soil moisture

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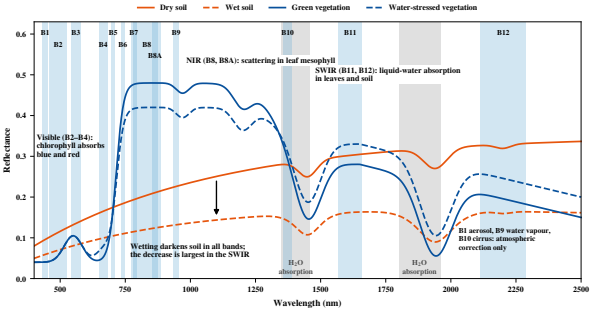
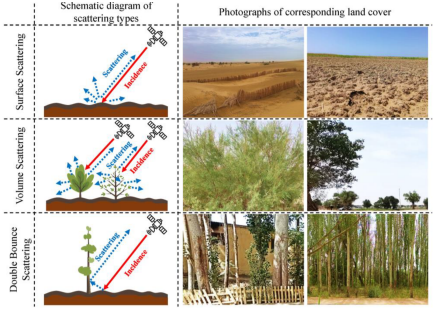
Motivation Different ways of obtaining soil moisture

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\text{SM}, \text{roughness}, \text{vegetation})$	top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST-vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	$\text{SM} = f_\theta(\text{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

Motivation Multimodal remote sensing



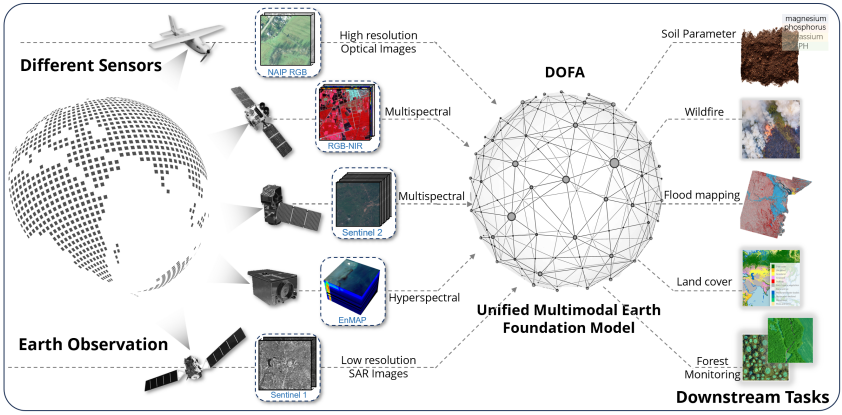
Optical: dry vs wet soil; green vs water-stressed vegetation



SAR scattering types (Nurmemet et al., 2025)

Soil and vegetation water change both optical reflectance and SAR backscatter

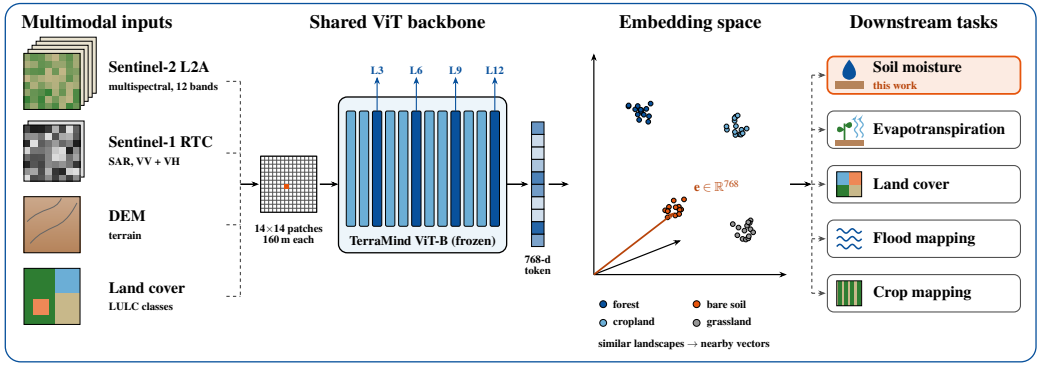
Motivation Geospatial foundation models (GFMs)



DOFA: one model for many sensors and many downstream tasks (Xiong et al., 2024, arXiv:2403.15356)

Once pre-trained on multimodal, unlabelled data, it works for many downstream tasks

Motivation TerraMind embeddings for soil moisture

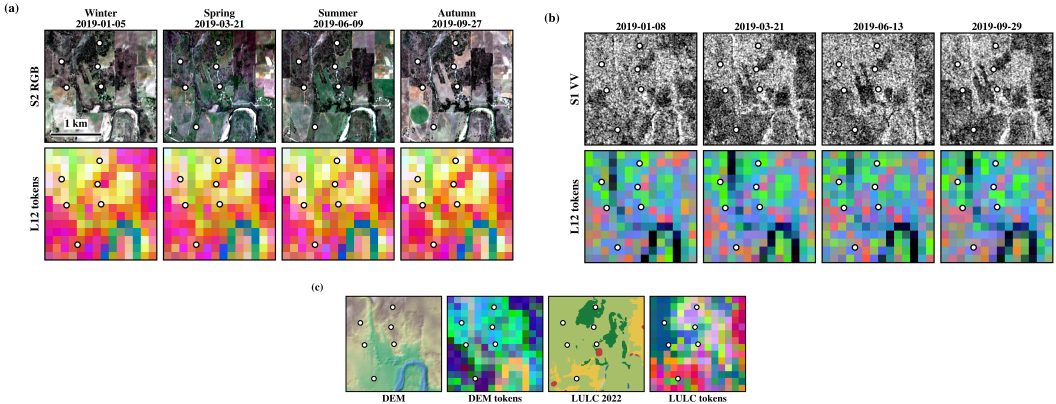


S2, S1, DEM and LULC → frozen TerraMind ViT-B → one 768-d embedding per 160 m patch → downstream tasks

The same embedding can serve many tasks; in this case, soil moisture

Motivation

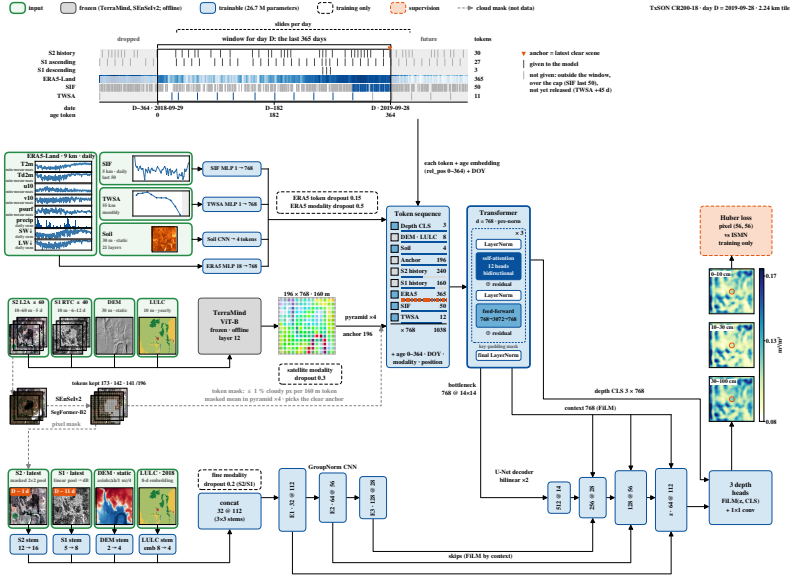
Example: TerraMind embeddings at one site (TxSON, 2019)



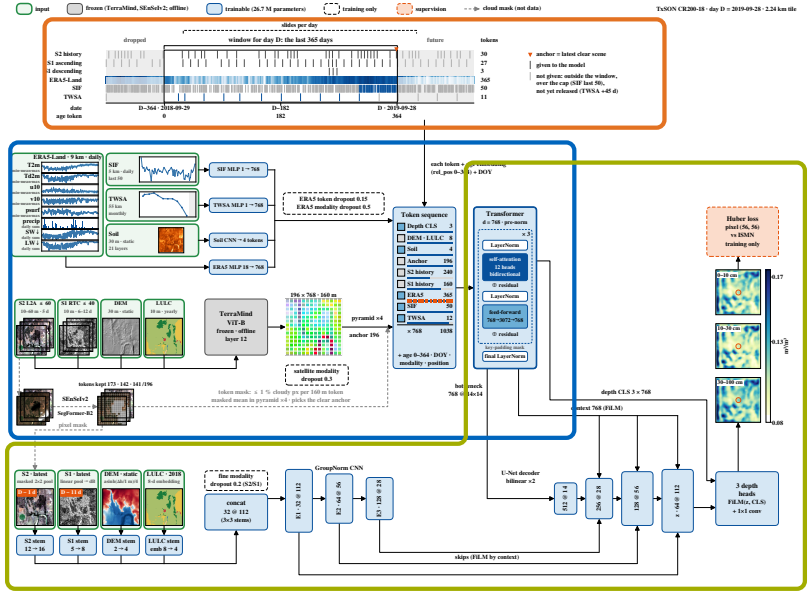
(a) S2 in four seasons, (b) S1 VV, (c) DEM and LULC, each with its TerraMind L12 tokens

Each 160 m patch → one 768-d vector; colours show its first 3 principal components

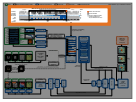
Method Model architecture

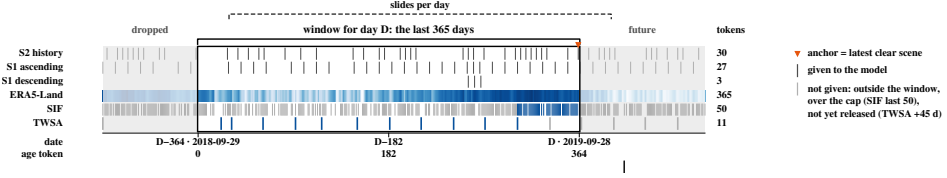


Method Model architecture

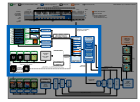


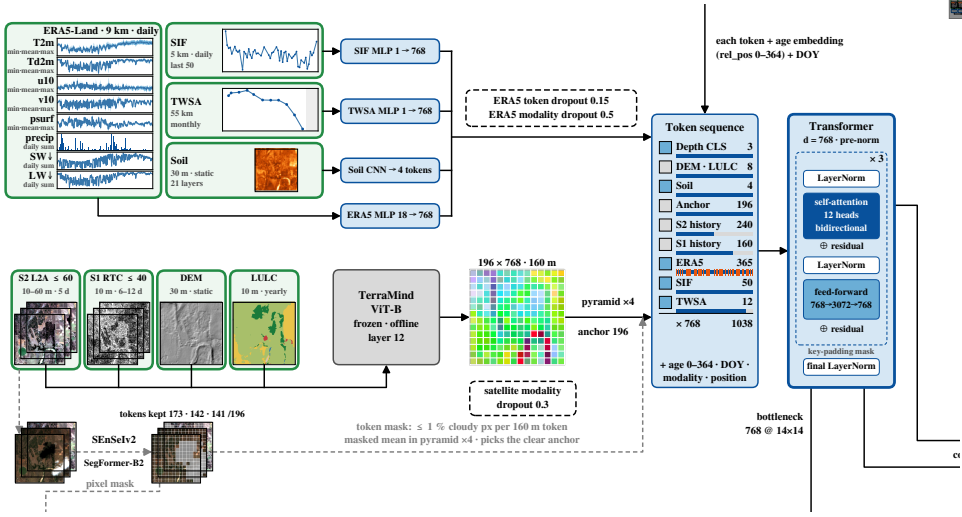
Method Inputs: what the model sees for day D



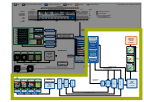


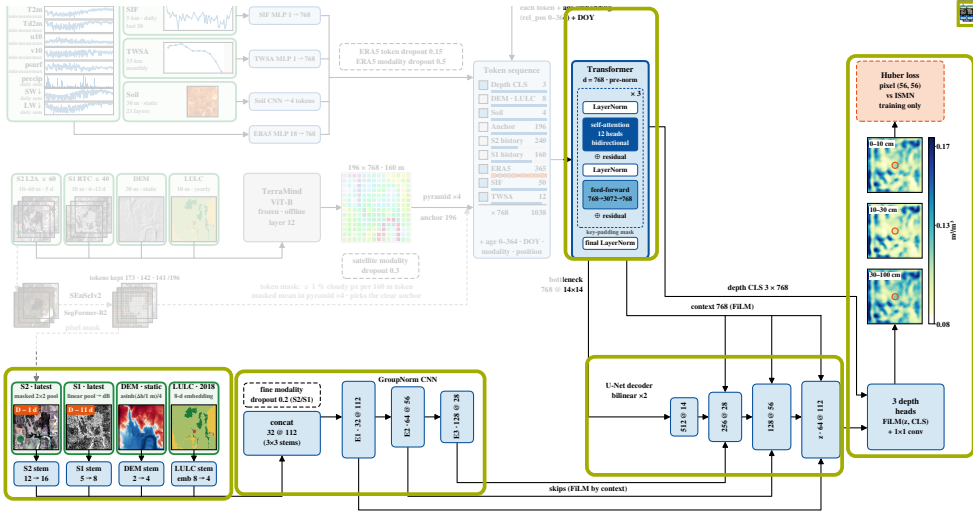
Method Coarse path: tokens → transformer



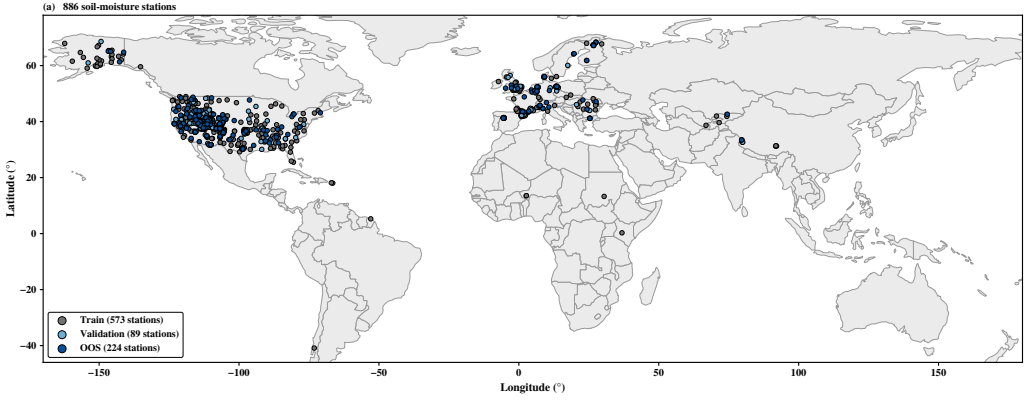


Method Fine path: 20 m CNN → U-Net decoder → depth heads





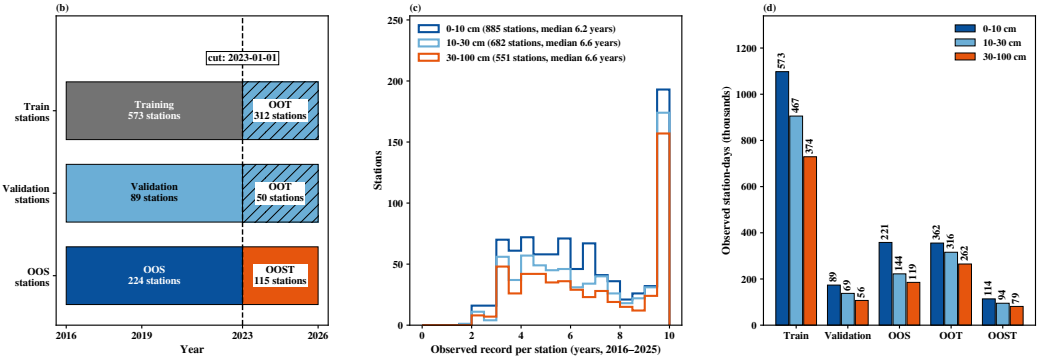
Data In-situ soil moisture: global coverage



886 soil-moisture stations (ISMN and flux sites): train 573, validation 89, out-of-sample (OOS) 224

Dense in North America and Europe; sparse in the tropics and the Southern Hemisphere

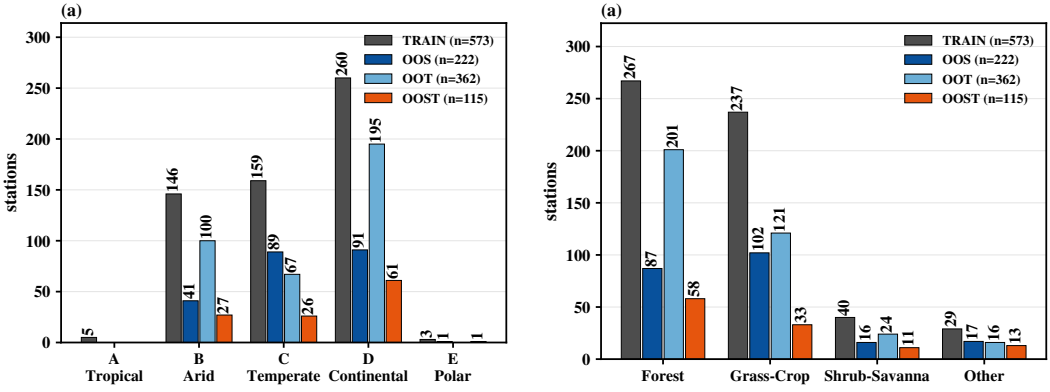
Data Depth coverage, record length and splits



(b) train / validation / OOS stations, time cut 2023-01-01; (c) observed record per station; (d) observed station-days

0–10 cm: 885 stations (median 6.2 yr); 10–30 cm: 682 (6.6 yr); 30–100 cm: 551 (6.6 yr)

Data Climate and land-cover coverage



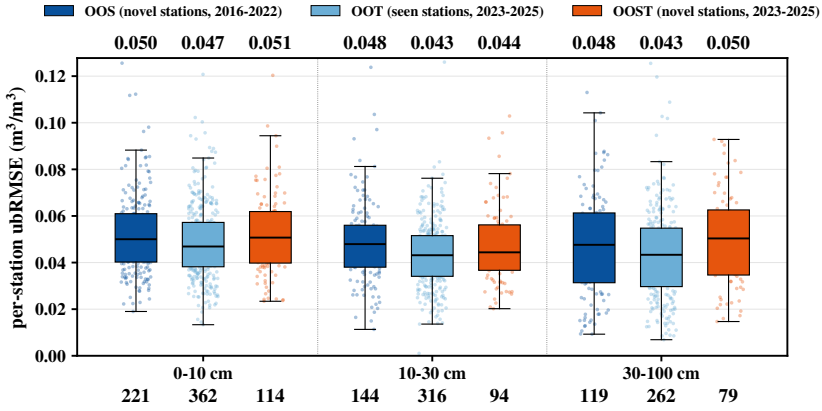
Stations per Köppen climate zone (left) and land-cover class (right), by split

Mostly continental, temperate and arid; forest and grass-crop; almost no tropics or polar

Stations	990 in inventory; **886** with soil moisture (train 573 / val 89 / OOS 224)
Depths	0–10 cm: 885 10–30 cm: 682 30–100 cm: 551 stations
Period	2016–2025; out-of-time test from 2023-01-01
Soil-moisture observations	**5.14 M** observed daily values (2.10 M / 1.68 M / 1.37 M per depth)
Sentinel-2 L2A	272,281 scenes (994 sites); cloud $< 75\,\%$ per scene, cloudy tokens masked
Sentinel-1 RTC	496,835 scenes: 251,310 ascending + 245,525 descending
ERA5-Land	18 variables, 2.46 M station-days (990 stations)
Static / other	DEM, LULC, soil (OpenLandMap), SIF, TWSA

About 5.1 million daily soil-moisture observations at 886 stations, 0–100 cm

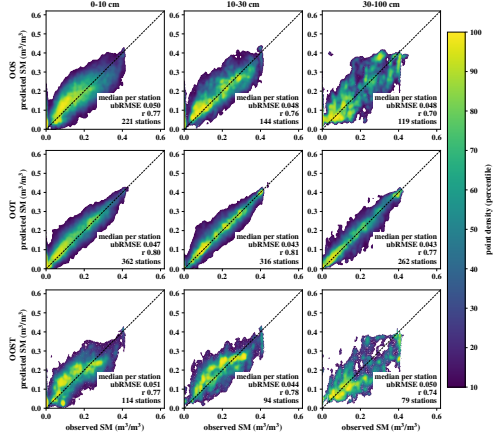
Results Per-station ubRMSE by depth



One point per station; medians above, station counts below

Median per-station ubRMSE 0.043–0.051 m^3/m^3 at every depth and split

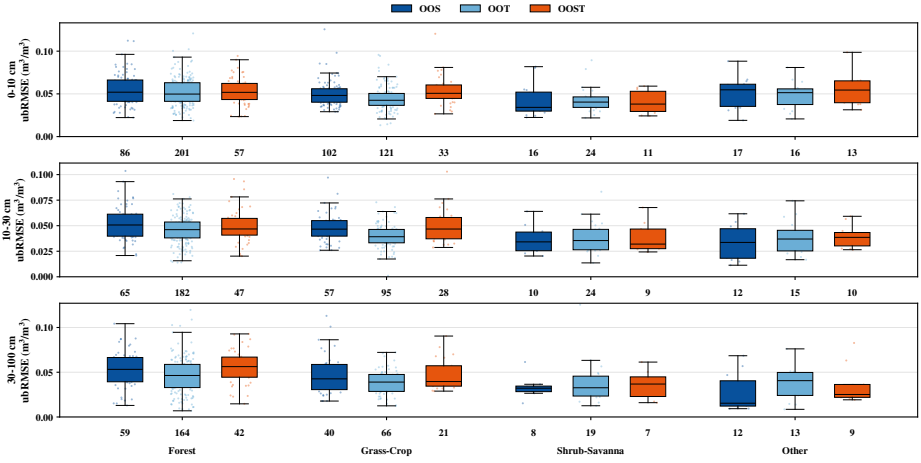
Results Predicted vs observed, by split and depth



Daily samples coloured by point density (sparsest 10 % not drawn); metrics = median over stations; rows OOS, OOT, OOST

New years at known sites fit far better than new sites

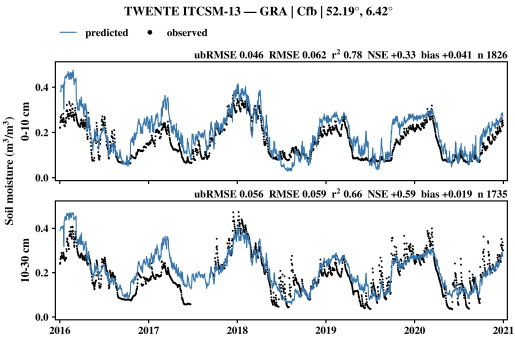
Results Per-station ubRMSE by land cover

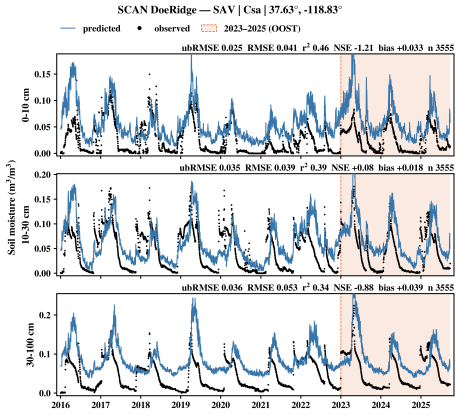


Rows: depths; groups: land-cover class; colours: OOS, OOT, OOST

Similar error across land cover; forest slightly highest, shrub-savanna lowest

Results Sites where the time series match



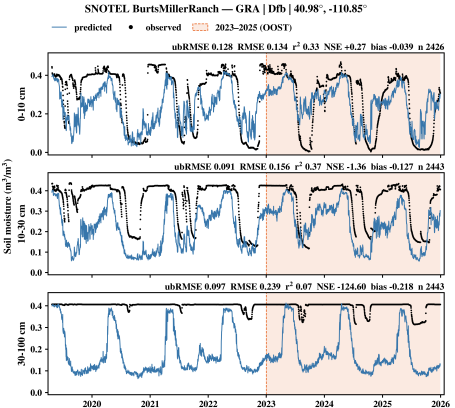


TWENTE ITCSM-13, Netherlands **SCAN Doe Ridge, California**

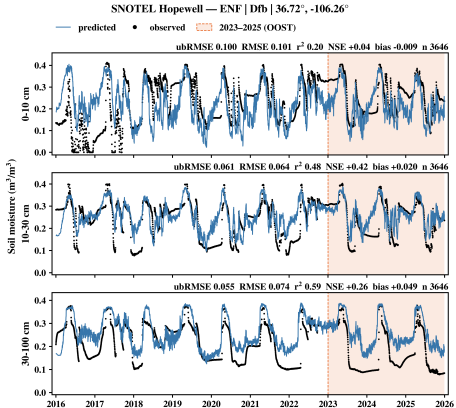
OOS stations; blue predicted, black observed; Twente has two depths, Doe Ridge three

Dynamics right at every depth (r 0.57–0.88, ubRMSE 0.024–0.056); level off by up to 0.04

Results Sites where the time series have some issues



SNOTEL Burts Miller Ranch, Utah

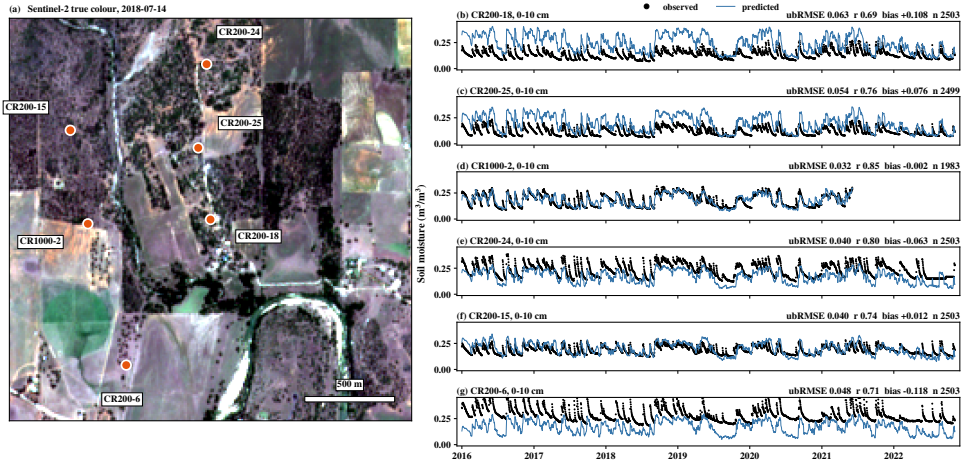


SNOTEL Hopewell, New Mexico

OOS stations observed at all three depths, highest ubRMSE; shaded = 2023–2025 (OOST)

Deep layer (30–100 cm) wrong in summer: too dry at Burts Miller, too wet at Hopewell

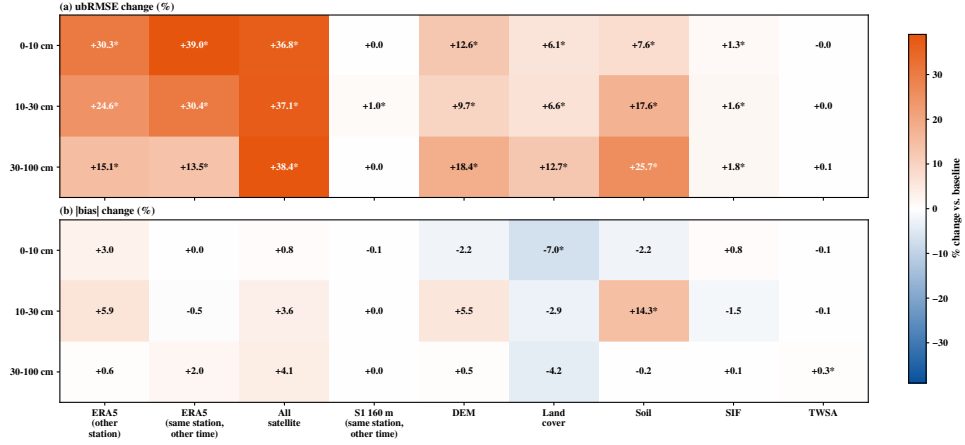
Results TxSON: six stations inside one 2.2 km tile



(a) Sentinel-2 true colour with the six stations; (b–g) 0–10 cm, predicted vs observed

Timing right at every station (r 0.69–0.85); offsets remain (bias −0.12 to +0.11)

Results Input ablation: which inputs does the model use?



*One input shuffled at a time; 221 OOS stations; * = significant*

ERA5 matters most at the surface (+30–39 % vs +14–15 % deep); satellite equally at all depths (+37–38 %); DEM, land cover and soil grow with depth

RQ1 Do frozen GFM embeddings add skill beyond a multimodal driver baseline?

Ans. **GFM embeddings carry a site-specific signal, comparable to climate.** Swapping satellite tokens between stations raises ubRMSE by 37–38 % at every depth (ERA5 from another station: +15–30 %); reordering S1 in time at the same station costs ≈0 %. Added skill still needs a drivers-only run.

RQ2 How does the model generalise to unseen sites?

Ans. **New sites are slightly worse than seen stations.** Median per-station ubRMSE 0.048–0.050 at new sites vs 0.043–0.047 at seen stations.

RQ3 How does skill vary within the same tile?

Ans. **Timing is captured, level may not be.** At TxSON, r is 0.69–0.85 and ubRMSE 0.032–0.063 at all six stations, while bias runs from −0.12 to +0.11 inside one 2.2 km tile.

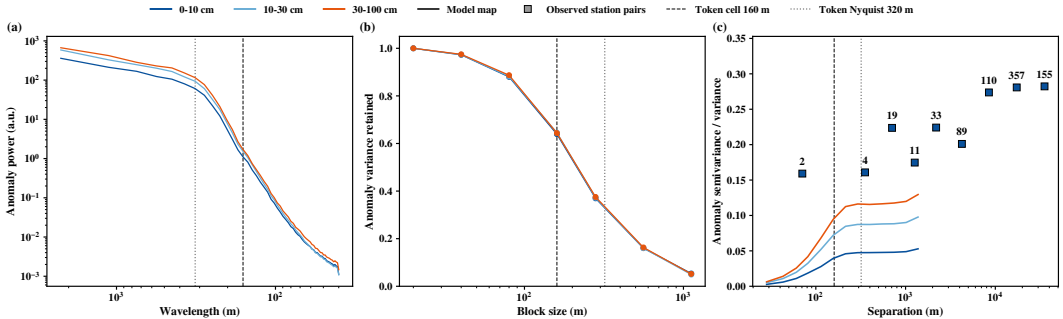
RQ4 Which modalities matter, at which depth?

Ans. **Depth changes what the model relies on.** ERA5 dominates the surface; satellite matters equally at all depths; soil, DEM and land cover gain weight at depth.

What I could not present

1. Effective resolution of the map 2. Where the station bias comes from 3. Voids in the density plot 4. Distribution of soil moisture

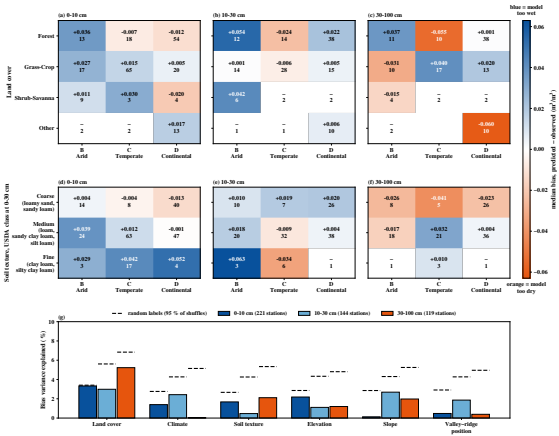
Other analysis 1. Effective resolution of the map (TxSON)



30 OOS TxSON tiles. (a) Spectrum of map anomalies; (b) variance kept after block averaging; (c) map variogram vs observed station pairs (numbers = pairs)

Map has structure down to 250 m

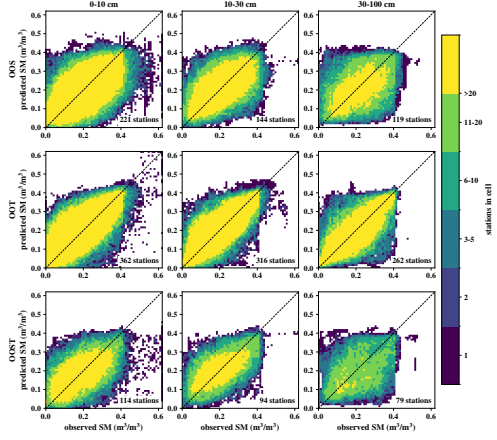
Other analysis 2. Where the station bias comes from



Median OOS bias by land cover, soil texture and climate; (g) bias variance explained vs random labels

Bias is not explained by land cover, climate, soil texture or terrain ($\leq$5 %, none above random)

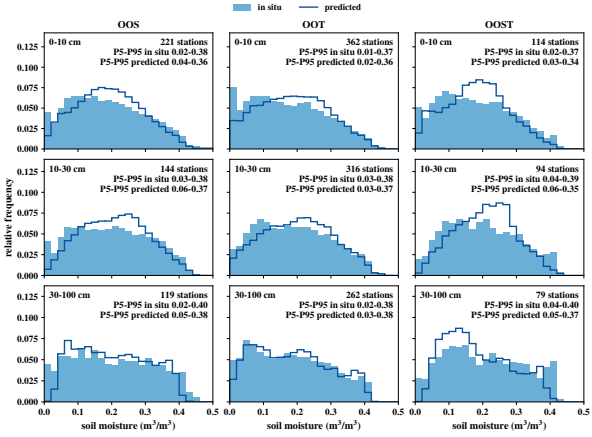
Other analysis 3. Voids in the density plot



Same grid as the density plot; colour = number of distinct stations with samples in each cell

Voids are not empty: many stations pass through briefly, a few long-record stations make the dense blobs

Other analysis 4. Distribution of soil moisture



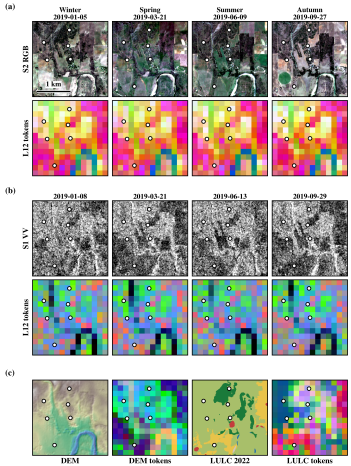
All daily samples; filled = in situ, line = predicted; box: P5-P95 range of each

Distribution matches; the model narrows the dry end

Eight research questions

Block	RQ	Question
Core	RQ1	Do frozen GFM embeddings add skill beyond a multimodal driver baseline?
	RQ2	What does the model generalise to — places or processes?
	RQ3	How does skill vary across the landscape?
Mechanism	RQ4	Which modalities matter, at which depth?
	RQ5	Does the fine-resolution path carry real sub-km information?
Negative	RQ6	Is land-surface temperature a useful auxiliary signal for SM?
Outlook	RQ7	Can the model detect management signals (irrigation)?
	RQ8	Does the representation transfer to an unseen region?

RQ1 Do frozen GFM embeddings add skill beyond a multimodal driver baseline?



TerraMind L12 tokens for S2, S1, DEM and LULC at one tile (TxSON, 2019).

Have

- Modality-shuffle ablation, §59 model, 221 OOS stations (next slide)
- Satellite tokens swapped across stations: +37 % ubRMSE, r 0.77 $\rightarrow$ 0.52

Missing

- Drivers-only model (10 epochs)
- Raw-imagery control (GFM vs pixels)
- Within-station S2 shuffle

RQ1 What does the model use the GFM embeddings for? (shuffle ablation)

	0–10 cm		10–30 cm		30–100 cm	
Shuffled input	ΔubRMSE (r)	worse	ΔubRMSE (r)	worse	ΔubRMSE (r)	worse
All satellite, cross	+36.8% (0.52)	97%	+37.1% (0.52)	96%	+38.4% (0.44)	92%
S2, cross	+4.9% (0.74)	84%	+6.8% (0.72)	87%	+8.7% (0.63)	85%
S1, cross	+4.8% (0.73)	82%	+7.7% (0.70)	84%	+14.4% (0.58)	86%
S1, within	0.0% (0.75)	49%	+1.0% (0.73)	59%	0.0% (0.64)	49%
DEM, cross	+12.6% (0.71)	90%	+9.7% (0.68)	86%	+18.4% (0.59)	96%
LULC, cross	+6.1% (0.73)	82%	+6.6% (0.70)	82%	+12.7% (0.62)	87%
Fine 20 m path, cross	+0.5% (0.75)	56%	+1.3% (0.74)	65%	+0.1% (0.68)	52%
ERA5, cross	+30.3% (0.50)	95%	+24.6% (0.54)	91%	+15.1% (0.50)	86%
ERA5, within	+39.0% (0.35)	88%	+30.4% (0.37)	81%	+13.5% (0.34)	73%
Soil, cross	+7.6% (0.72)	87%	+17.6% (0.63)	92%	+25.7% (0.55)	94%
SIF, cross	+1.3% (0.75)	63%	+1.6% (0.73)	62%	+1.8% (0.67)	71%
TWSA, cross	0.0% (0.77)	50%	0.0% (0.76)	53%	+0.1% (0.71)	58%

Baseline: ubRMSE 0.050 / 0.048 / 0.048, r 0.766 / 0.761 / 0.702 (n = 221 / 144 / 119). *Cross* = another station's input (kills site identity); *within* = same station, shuffled dates (kills timing).

RQ1 What we can and cannot claim today

Can claim

1. Reliance on GFM tokens $\approx$ reliance on weather (+37 % vs +30 %).
2. Tokens are a *static landscape descriptor*: S1 within-station shuffle = 0 %.
3. They shape the response, not just the mean (ubRMSE is bias-free): *GFM embeddings condition the weather$\rightarrow$SM transfer on the landscape.*
4. Depth pattern inverted vs a moisture signal: S1 +4.8 % $\rightarrow$ +14.4 % with depth.
5. Modalities redundant: S2+S1+DEM+LULC $\approx$ 28 % $<$ 37 % joint.
6. Fine 20 m path and TWSA unused.

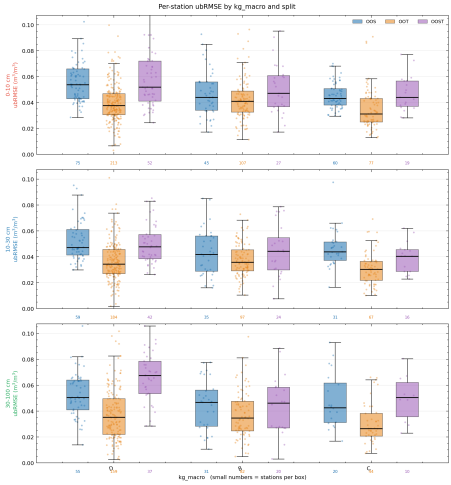
Cannot claim yet

- *Added* skill — a shuffle measures reliance
- GFM better than raw imagery
- Anything about S2 dynamics

To answer fully

- Within-station S2 shuffle (eval only)
- Run A: drivers + soil, no tokens
- Run B: A + static site descriptors
- Full $>$ B $>$ A $\Rightarrow$ dynamic skill;
  Full $\approx$ B $>$ A $\Rightarrow$ site info only

RQ2 What does the model generalise to — places or processes?



Per-station ubRMSE by climate zone and split (OOS / OOT / OOST).

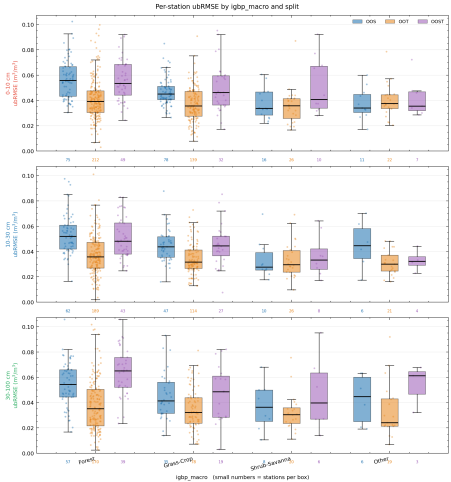
Have

- Splits §47: train 573 / val 89 / OOS 224; OOT 2023–2025
- OOT error is *lower* than OOS in every climate zone: new years at known sites are easy, new sites are hard
- `pw_stage2a_L3`: OOT 0.0434 beat val 0.0507
- Step-0 offsets probe: pair sign 0.50; site-fingerprint verdict confirmed (27572594)

Missing

- One val / OOT / OOS table for the selected model

RQ3 How does skill vary across the landscape?



Per-station ubRMSE by land-cover class and split.

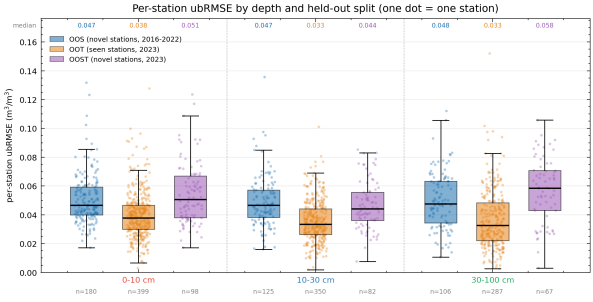
Have

- Per-station metrics by climate zone and land cover
- LULC, elevation band, soil, HAND for every station
- HAND stable under DEM noise; TWI is not — use HAND

Missing

- Error vs aridity, HAND, soil texture, per depth
- Flag elevation sentinels first (18 REMEDHUS = −99.9, Baytik = 0)

RQ4 Which modalities matter, at which depth?



Per-station ubRMSE by depth.

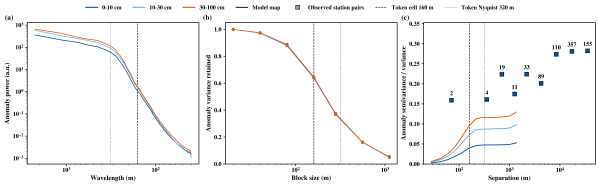
Have (shuffle ablation, RQ1 table)

- ERA5 dynamics dominate the surface (+39 %) and fade with depth (+14 %)
- Soil properties grow with depth (+7.6 % → +25.7 %)
- SIF ≈ 1.5 %; TWSA ≈ 0
- §58: 180-day ERA5 window

Hypothesis status

- "TWSA drives the root zone" — not supported so far

RQ5 Does the fine-resolution path carry real sub-kilometre information?



Effective-resolution test at TxSON (§70).

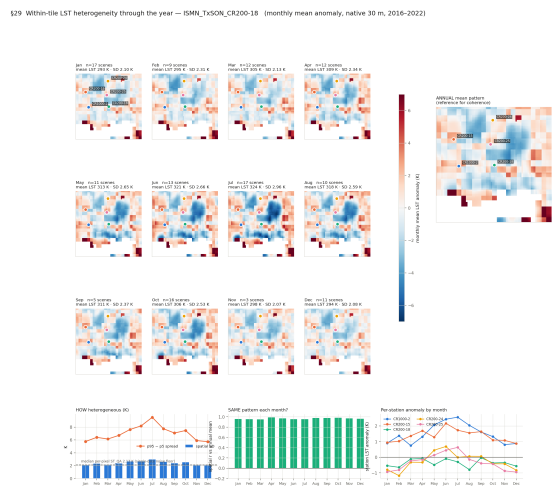
Have

- §53 fine ablation: SM ignores the 20 m path (LST reads it)
- Fine-path shuffle: +0.1 to +1.3 % ubRMSE
- §48 map flattens after epoch 8
- Overijssel grid run (§68, paused)

Framing

- A publishable negative result if framed well

RQ6 Is land-surface temperature a useful auxiliary signal for soil moisture?



Within-tile Landsat LST anomaly by month, TxSON CR200-18 (§29).

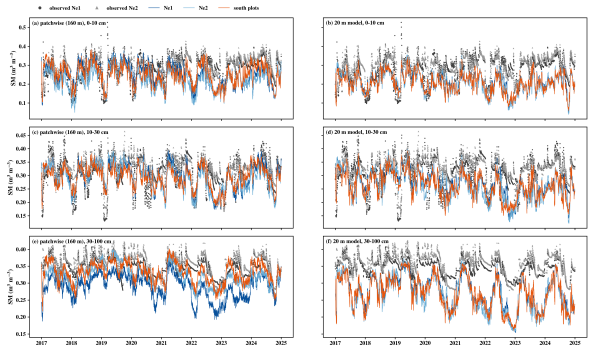
Have (all negative)

- Within-tile LST pattern is static: $r = +0.967$ across months
- Simpson's paradox: +0.167 pooled, −0.077 within station
- ECOSTRESS DTR = day pattern (r = +0.816); G0 fails
- In training LST hurts SM: val 0.0537 with vs 0.0487 without

Status

- Ready to write as a short paper / section

RQ7 Can the model detect management signals such as irrigation?



Mead Ne1/Ne2/Ne3: model vs observed SM, 2017–2024 (§69).

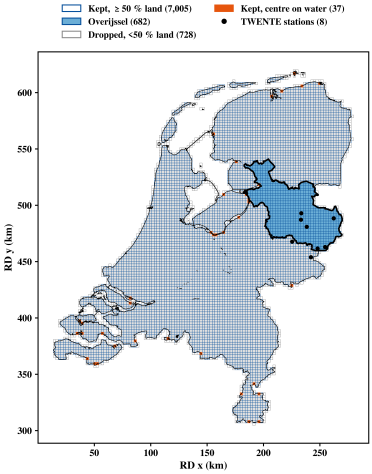
Have

- Mead daily SM built for Ne1/Ne2/Ne3
- Irrigation visible only in the dry year (2022)

Missing

- Irrigation log covers 2014–2021 only
- Prediction-minus-observation analysis vs irrigation dates

RQ8 Does the representation transfer to an unseen region?



Overijssel grid tiles (§68).

Have

- Overijssel: cloud mask done 688/688
- TWENTE: 8 stations, all OOS (end 2020, no OOT)

Missing

- Prep → tokenise → post → predict
- Map + TWENTE station comparison